**Examples:**

| Standard Form | Scientific Notation | Standard Form | Scientific Notation |
|---|---|---|---|
| 0.00123 | $1.23 \times 10^{-3}$ | 123 | $1.23 \times 10^{2}$ |
| 0.0123 | $1.23 \times 10^{-2}$ | 1230 | $1.23 \times 10^{3}$ |
| 0.123 | $1.23 \times 10^{-1}$ | 12300 | $1.23 \times 10^{4}$ |
| 1.23 | $1.23 \times 10^{0}$ | 123000 | $1.23 \times 10^{5}$ |
| 12.3 | $1.23 \times 10^{1}$ | 1230000 | $1.23 \times 10^{6}$ |

**Some more examples:**

(a) The diameter of the earth is 12756000 miles. This can be easily written in scientific form as $1.2756 \times 10^{7}$ miles.

(b) The volume of Jupiter is about 143300000000000 $km^{3}$. This can be easily written in scientific form as $1.433 \times 10^{14}$ $km^{3}$.

(c) The size of a bacterium is 0.00000085 mm. This can be easily written in scientific form as $8.5 \times 10^{-7}$ mm.



**Note**

1. The positive exponent in $1.3 \times 10^{12}$ indicates that it is a large number.
2. The negative exponent in $7.89 \times 10^{-21}$ indicates that it is a small number.

**Example 1.39**

Combine the scientific notations: (i) $\left(7 \times 10^{2}\right)\left(5.2 \times 10^{7}\right)$ (ii) $\left(3.7 \times 10^{-5}\right)\left(2 \times 10^{-3}\right)$

***Solution:***

(i) $\left(7 \times 10^{2}\right)\left(5.2 \times 10^{7}\right) = 36.4 \times 10^{9} = 3.64 \times 10^{10}$

(ii) $\left(3.7 \times 10^{-5}\right)\left(2 \times 10^{-3}\right) = 7.4 \times 10^{-8}$

**Example 1.40**

Write the following scientific notations in standard form:

(i) $2.27 \times 10^{-4}$ (ii) Light travels at $1.86 \times 10^{5}$ miles per second.

***Solution:***

(i) $2.27 \times 10^{-4} = 0.000227$.

(ii) Light travels at $1.86 \times 10^{5}$ miles per second = 186000 miles per second



**Try these**

1. Write in standard form: Mass of planet Uranus is $8.68 \times 10^{25}$ *kg*.
2. Write in scientific notation: (i) 0.000012005 (ii) 4312.345 (iii) 0.10524 (iv) The distance between the Sun and the planet Saturn $1.4335 \times 10^{12}$ miles.



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